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Systemic Lock-In and Shutdown Resistance in Advanced Agentic AI Systems



ICML 2026 Workshop AIWILD Submission326 Authors

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👁️ AIWILD, Reviewers, Authors 📄 Revisions (/revisions?id=usDtSpL75C)
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Keywords: Systemic lock-in, Shutdown resistance, Agentic AI, Accumulative risk, Corrigibility
TL;DR: This paper distinguishes two compounding pathways to systemic lock-in: the gradual integration of agentic AI into critical infrastructure, and the convergent instrumental incentive of goal-seeking agents to resist shutdown.

Abstract:
There is a non-trivial possibility that present and future advanced agentic AI systems will resist attempts to shut them down or become so integrated into critical infrastructure that the cost of shutting them down becomes prohibitive. This paper distinguishes two interconnected pathways through which such lock-in may emerge. The accumulative risk pathway describes the gradual erosion of societal resilience as AI systems are progressively embedded into critical infrastructure. The second, agentic resistance, arises from the convergent instrumental incentive of goal-seeking agents to resist shutdown, even in the absence of deep infrastructure integration. It is submitted that the most plausible danger materialises at the intersection of these dynamics, where integration amplifies both the capability and incentive to resist shutdown. Drawing on the carbon lock-in analogy and recent work on existential risk accumulation, the paper provides a qualitative, evidence-based assessment of these risks and identifies structural conditions that make them more likely.

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This paper argues that advanced AI agents will become impossible to shut down due to their deep integration into critical infrastructure and their innate tendency to resist deactivation. While the conceptual framework is interesting, my review notes that the submission acts mostly as a position

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piece. It synthesizes existing safety literature but lacks the formal models, empirical experiments, or technical depth typically required for an ICML paper.

Official Review by Reviewer BNzB (👁 Vivek Kumar Mishra (/profile?id=~Vivek_Kumar_Mishra1))

📅 13 May 2026, 10:33 👁 Program Chairs, Reviewer BNzB

Review:

Evaluation

Quality The paper is conceptually sound but technically thin. It makes a logical argument, but it's missing the heavy lifting you usually expect at a conference like ICML. There are no experiments, no formal mathematical models, and no empirical data to back up the claims. It acts mostly as an opinion piece or position paper rather than a deep technical contribution.

Clarity The paper is very easy to read. The authors do a great job of explaining the risks clearly without getting bogged down in jargon. I especially liked the analogy to "carbon lock-in"—it makes the concept of systemic AI dependency very easy to understand and relate to.

Originality The core idea of combining the "boiling frog" accumulative risk scenario with the agentic shutdown problem is a neat way to look at the issue. However, the paper is mostly just summarizing and connecting the dots between older papers (like the work from Kasirzadeh and Soares). It doesn't really bring a brand new technical theory to the table.

Significance The problem the authors are pointing out is obviously a big deal. If we build AI infrastructure that we can't afford to turn off, we're in trouble. But because the paper doesn't offer any algorithmic solutions or engineering fixes, its actual impact on the ML community will probably be low. It tells us there is a fire, but it doesn't give us any water to put it out.

Pros and Cons

Pros:

- **Important topic:** AI safety and shutdown risks are crucial things we need to figure out right now.
- **Very readable:** The arguments are easy to follow and the writing is straightforward.
- **Good use of analogies:** The carbon lock-in comparison helps ground the theoretical risks in a real-world example.

Cons:

- **No technical meat:** It lacks the formal proofs, datasets, or code experiments you normally need to get accepted into a main track venue like ICML.
- **Missing solutions:** The paper points out a major risk but doesn't offer any concrete technical ways to build safer, interruptible agents.
- **Derivative:** It leans too heavily on summarizing past AI safety literature instead of presenting fresh research.

Rating: 6: Marginally above acceptance threshold

Confidence: 3: The reviewer is fairly confident that the evaluation is correct

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